

Supplementary Information

How much of a nervous-system model does behaviour identify?

Chenxi He, Cavendish Laboratory, University of Cambridge
ch2067@cam.ac.uk

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This Supplementary Information provides the extended methods and the complete per-experiment quantities underlying each main-text claim. It comprises: the measurement methods and their controls; the provenance of the external biological datasets; the confirmatory experiments that extend the probes to higher parameter dimension, to named single channels, to the full connectome, to the internal neural state, and to a fifth simulator; and the experiments that identify the manifold with the eigenworm space, establish its complementary tiling by behaviour, and demonstrate the structure across three phyla, together with one inconclusive control that is reported in full. Every quantity is drawn from a saved result file, named per section, and is regenerated from that file by the accompanying reproduction code.

1 Measuring the manifold: methods

1.1 Multistart parameter drift (S1.1)

For each simulator we run N independent optimisations (separable CMA-ES or SPSA) from independent starts against one *fixed* behavioural target, (Table 1) and record (i) behavioural recovery (fraction of the random-to-target behaviour-distance closed), (ii) the normalised per-dimension dispersion of the N recovered parameter vectors, and (iii) the PCA participation ratio of the solution cloud. A behaviourally-converged but parameter-dispersed cloud is direct evidence of a positive-dimensional equivalence manifold.

Table 1: **Fixed-target multistart optimisation on two simulators.** Each row reports N independent optimisations started from independent points and fitted to a single fixed behavioural target. dim: number of free parameters; N : number of independent runs; success: runs reaching the behavioural target; behav. rec.: median fraction of the random-to-target behavioural distance closed; θ disp.: dispersion of the recovered parameter vectors, normalised per dimension; cloud dim: PCA participation dimension of the solution cloud. A solution-cloud dimension is not by itself evidence of a manifold. Run under the same optimiser and budget, a strictly convex problem with a unique optimum and no manifold at all gives a cloud of 20.2 of 126 dimensions, against the 21.7 measured here, because a participation dimension describes the shape of the scatter and not its size. The quantity that does separate the two is the magnitude of the dispersion, which differs by a factor of 78 between the convex control and a construction with twenty flat directions. Optimisers are separable CMA-ES (FlyGym) and SPSA (BAAIWorm); the flybody objective overfits per seed without a prior and is therefore excluded here and reported only in the curvature analysis.

sim / method	dim	N	success	behav. rec.	θ disp.	cloud dim
FlyGym / CMA-ES	126	30	29/30	70.8%	0.098	21.7/126
BAAIWorm / SPSA	5	10	5–7/10	—	0.67	—

The FlyGym 126-d cloud occupies an effective 21.7 of 126 dimensions. (The flybody 515-d objective overfits per-seed without a prior and is excluded from the drift panel; it appears only in the Hessian

analysis.) On the full coupled **real modWorm** model (OpenWorm connectome, 279 neurons), the manifold is read from the per-mechanism Hessian (S1.2): only 2 of 7 biophysical-mechanism directions carry behavioural curvature, the rest forming the flat manifold. (A direct random-perturbation probe gives the same qualitative picture but with an assay-dependent magnitude, so we report the manifold by its effective dimension, not by the raw sloppiness ratio, which is not comparable across simulators with different observables.) File `modworm-REAL_manifold.json`.

1.2 Hessian eigenspectrum (S1.2)

At each converged optimum we form the behavioural-loss Hessian by finite differences (low dimension) or a Lanczos approximation (high dimension) and report the effective dimension (number of eigendirections carrying 90%/99% of the spectral mass) and the count of near-flat directions ($|\lambda|$ at the finite-difference floor). On BAAIWorm, over 5 mechanism-class scalings at $\varepsilon=5 \times 10^{-3}$, the eigenvalues are $\lambda_1=+348$, $\lambda_2=-340$ and $\lambda_3=-214$ (a saddle), with the ion and passive eigendirections flat to machine precision over a parameter L2 length of 2.0; the sign structure is stable across $\varepsilon \in [10^{-4}, 5 \times 10^{-2}]$. On the 767-parameter flybody policy (Lanczos $m=40$, $\varepsilon=0.005$, anchor $L_0=9.513$) there are six positive stiff eigenvalues (8.2×10^4 down to 2.1×10^3) and 28 of 40 sampled directions are flat ($|\lambda| \sim 10^{-11}$), giving effective dimension 6 at 90% and 11 at 99% with participation ratio 3.76. On modWorm, the per-mechanism Gauss–Newton Hessian of the full coupled neural+body model on the OpenWorm connectome (1, 2) (279 neurons, 7 biophysical mechanisms) has effective dimension 2 of 7 at both thresholds with participation ratio 1.29; the stiff mechanisms are the synaptic and gap-junction conductances and the synaptic rise time, while the membrane capacitance and the sigmoid gain carry zero elasticity, again placing connectivity above single-cell intrinsics. File `modworm-REAL_b1.json`.

1.3 Finite-difference validity across simulators (S1.5)

A Gauss–Newton curvature assembled by central differences is meaningful only where the response to a perturbation rises above the numerical floor and remains close to proportional to the step. That condition is not automatic, and we tested it in every simulator before reporting any curvature. Where it fails, the failure is a property of the simulator rather than of the organism, and separating the two is part of what this measurement is for.

BAAIWorm. Scaling one connection weight, the whitened response is 1.43×10^{-5} , 2.84×10^{-5} , 7.10×10^{-5} , 1.42×10^{-4} and 2.84×10^{-4} at steps of 0.05, 0.1, 0.25, 0.5 and 1.0: the ratio to the step is constant at 2.84×10^{-4} to four significant figures across a twentyfold range. Repeat simulations at identical parameters are bit-identical, so the noise floor is exactly zero and no part of the measured curvature is finite-difference noise.

flyvis. The response is proportional to the step across two decades (0.005 to 0.5), with ratio 0.498 for the per-cell-type biases and 0.241 for the time constants, rising only slightly for the synaptic strengths.

FlyGym. A floor of 0.037 in the whitened joint-angle response is present already at a step of 0.001 and does not fall further, so at the step of 0.05 used in the original probe roughly three quarters of the measured difference is stochasticity of the controller rather than parameter sensitivity. Proportionality is recovered only above a step of about 0.25. The accompanying null, in which columns are built from seed-to-seed variation at identical parameters, gives an effective dimension of 7 (90%) / 22 (99%), which is the floor against which the 16 / 38 measured with 4200 observables should be read.

Prinz–Marder pyloric circuit. Trajectory-level observables cannot be used here. Changing a

single conductance by 0.1% already decorrelates the membrane-potential traces (Pearson $r = 0.42$; $r = 0.06$ at 25%), because any parameter change shifts the phase of the rhythm and a pointwise comparison of two phase-shifted oscillations saturates. The whitened difference is therefore independent of the step, the Jacobian scales as $1/\delta$, and a step-size convergence test cannot detect the problem: the normalised spectrum, and hence the effective dimension, is unchanged at 25.4, 25.7 and 25.7 for steps of 0.1, 0.05 and 0.025. The phase-invariant summary statistics used in this literature are the correct response to this, not a convenience, and we report the pyloric circuit through them.

modWorm. The posture response is discontinuous in the parameters. Scaling the synaptic rise time constant changes the body angles by 4×10^{-9} at +0.1% and 3×10^{-8} at +1% (numerically zero), then by 0.141 at +5% and by the same 0.141 at +10%. Three of the seven mechanism classes, the leak conductance, the membrane capacitance and the sigmoid gain, change the posture by 10^{-12} to 10^{-7} at any step, that is, not at all: they are unconnected in this configuration rather than behaviourally sloppy, so the model has four mechanisms with a measurable effect and not seven. We report modWorm results with both caveats attached and do not use it for any conclusion that a better-conditioned model can support.

The general point is that an effective dimension is a statement about a measurement as much as about a model, and it should be accompanied by the step-size scan, the noise floor and the count of parameters that do anything at all. Files `E44_baai_fd_controls.json`, `E33_stg_fd_convergence.json`, `E1_flygym_null.json`.

1.4 Simulation-based calibration as a manifold probe (S1.3)

An amortised neural posterior (masked autoregressive flow (3), `sbi` (4)) is trained once on the prior, and per-dimension rank-based SBC (5, 6) is run (M/L posterior/rank draws); “pass” is a per-axis Kolmogorov–Smirnov test at $\alpha=0.05$. Pass counts: real modWorm 7/7 ($N=300$ coupled-model sims), FlyGym 22/48, BAAIWorm full5 5/5. FlyGym posterior-mean gain rel-error 0.018. The partial pass on FlyGym delimits the calibratable subspace of the manifold; the full pass on the lower-dimensional real-modWorm and BAAIWorm probes reflects a correctly recovered posterior.

1.5 B1 — per-mechanism stiff/sloppy (S1.4)

A preliminary probe scaled five global mechanism classes and evaluated the raw finite-difference Hessian rather than its Gauss–Newton form. The per-parameter curvature magnitudes ordered the classes as synaptic 331.97, gap-junction 299.64, motor-output 270.50 (stiff) against ion-channel and passive-membrane ≈ 0 (sloppy), but the raw Hessian at that point carried mixed-sign eigenvalues (+348, −340, −214, 0, 0), indicating a saddle rather than a minimum. Curvature signs are therefore not interpretable at that point, and we do not use this probe for any reported result. It is superseded by the positive semi-definite Gauss–Newton decomposition over 20 named mechanisms (§5.2), which resolves the same classes at finer granularity, and by the Gauss–Newton analyses of every other system reported here. We retain it only to document the ordering it produced, which agrees with the Gauss–Newton result. File: `zerocompute_crosscheck.json`.

2 External references (no simulator used)

2.1 E1 — Prinz–Marder pyloric population (S2.1)

We re-gate the published conductance-based STG model population (7) to the canonical triphasic pyloric rhythm (all three cells bursting; period $\in [500, 2000]$ ms; strict AB/PD<LP<PY phase ordering; valid rate 0.56%) and probe valid points with the *same* Hessian-eigenspectrum method as S1.2, providing a direct comparison to B1. Over $K=8$ valid points: participation effective dimension 2.13/31; spectrum spanning ~ 10 orders of magnitude; 2–6 stiff directions; stiff = leak and inhibitory-synapse conductances, sloppy = fast Na conductances, consistent with the compensation and co-regulation documented for this circuit (8–11). An expanded sweep confirms this at larger sample: 40,000 prior draws yield 151 rhythm-valid points, of which 60 were probed; through the 15 summary statistics the effective dimension is 2.4 ± 0.9 (range 1–4) across the points with complete statistics, matching the original estimate, and it is stable point to point. The same expansion also quantifies why raw voltage traces are not a usable observable for this oscillator: read through 6,600 trace points the apparent dimension is 25.4 ± 0.9 of 31 at every probed point, but the underlying finite differences fail their own convergence test—a 0.1% conductance change already decorrelates the trace (S1.5)—so the trace-based figure measures phase sensitivity rather than identifiable structure, the oscillator counterpart of the pointwise-loss failure documented at the cell scale (§7.1). Files: `e1b_results/e1b_hessian.json`, `e1b_results/e1b_gate.json`, `E3_pyloric_rank.json`, `E33_stg_fd_convergence.json`.

2.2 E2 — real neuron variability (S2.2)

Allen Cell Types electrophysiology, $n=1914$ mouse+human cortical neurons: per-property coefficient of variation 0.08–1.85 (stiff: resting potential 0.08, fast-trough 0.12; sloppy: adaptation 1.85, firing rate 1.11); property-space PCA participation dimension 4.78/11. File: `E2_allen_variability.json`.

2.3 E4 — ion-channel-gene conservation (S2.3)

40 *C. elegans* ion-channel/synaptic genes, mean ortholog %identity across *Caenorhabditis* (Worm-Base ParaSite / Ensembl Metazoa): range 77.94–99.91 (median 91.20); most conserved `unc-9` (gap junction, 99.91), `egl-19` (L-type Ca, 97.19); least `unc-49` (GABA-A, 77.94). File: `E4_channel_conservation.json`

2.4 E5 — isogenic behavioural manifold (S2.4)

OpenWorm Movement Database, $n=61$ isogenic N2 worms, 683 Tierpsy features: cross-individual participation dimension 9.75; 24 PCs for 90% variance, 50 for 99% (PC1 26.6%, PC2 10.7%). File: `E5_n2_manifold.json`.

3 Unified same-region probe test (result)

Computing the probes on the full coupled modWorm model in one shared prior region (per-mechanism log-multipliers, prior scale 0.15). **The two geometric probes agree:** per-mechanism Hessian stiffness versus negative manifold drift give Spearman $\rho=0.96$ ($p=5 \times 10^{-4}$; 7 mechanisms). **The amortised posterior is a calibration control, not a third ranking:** an NPE/MAF (`sbi`) trained on $N=300$ real modWorm simulations is well-calibrated on all 7 mechanisms (per-axis SBC 7/7, Kolmogorov–Smirnov $p > 0.05$), confirming the inference is sound; because a calibrated posterior calibrates every axis, SBC serves as a control rather than a per-axis identifiability

ranking. We therefore report a two-probe geometric agreement plus a calibration control. Files: `modworm_REAL_sbc.json`, `modworm_REAL_b1.json`.

4 Confirmatory experiments

These measurements extend the picture above to higher parameter dimension, to named single channels, to the full connectome, and to the internal neural state. Behavioural curvature is reported as the Gauss–Newton Hessian $H=J^\top WJ$ formed from the observable Jacobian $J_{ik}=\partial \text{obs}_i/\partial \theta_k$ (whitened by per-observable scale W), the Fisher-information form that quantifies which parameter directions behaviour constrains.

4.1 Higher-dimensional Hessian: FlyGym 48-d (S5.1)

The behavioural Hessian over the 48 actuator gains of the FlyGym walking controller has effective dimension 3 (90%) / 6 (99%) of 48 (participation ratio 2.15), with an eigenspectrum spanning 19.5 orders of magnitude (leading eigenvalues 20.4, 5.5, 2.7, 1.2, 0.80, 0.62, then < 0.06). That figure is set by the observable count: with ten summary scalars the Hessian can have rank at most ten, and it has exactly ten non-zero eigenvalues. Re-measured against 4200 joint-angle observables the effective dimension is 16 (90%) / 38 (99%), against a noise floor of 7 / 22 obtained from seed-to-seed variation at fixed parameters. The reported spectral span of 19.5 orders likewise reaches eigenvalues at 10^{-19} , below the stated finite-difference floor; above the floor the span is 3.85 orders. File `flygym_hessian48.json`.

4.2 Per-channel stiff/sloppy decomposition (S5.2)

Replacing the five global mechanism-class scalings with 20 named mechanisms (16 individual ion-channel conductances, the passive leak, the synaptic and gap-junction weights, and the motor output), the per-channel behavioural elasticity ($d \ln \text{obs} / d \ln \theta$) is measured under two observable sets. Through the full motor-neuron trajectory (10,320 observables; the assay the main text reports, file `E45_baai_named_channels.json`) the effective dimension is 3 (90%) / 6 (99%) over the twelve mechanisms with a measurable effect, with the inward rectifier IRK (123.5), the sodium leak NCA (117.7) and the potassium conductances EGL-36 (86.0) and EGL-2 (44.6) stiffest, and SLO-1/SLO-2 producing no measurable response. Through six summary kinematics the same decomposition gives 2 (90%) / 3 (99%) of 20 (eigenvalues 43.2, 5.84, 0.88, 0.47, remainder ≈ 0): the coarser assay reports fewer constrained directions, as the rank analysis predicts, while the stiff/sloppy identities agree. The stiff mechanisms are the passive leak, the gap-junction and synaptic weights, the motor output, and a few potassium conductances (EGL-36, EGL-2, IRK); the sloppiest are the fast voltage-gated calcium channels EGL-19 (elasticity 0.49) and UNC-2 (0.31) and the calcium-activated potassium channels SLO-1/SLO-2 (0.01–0.16). Behaviour is set by wiring, passive properties and a subset of potassium channels, and is nearly invariant to the fast calcium and calcium-activated potassium conductances, the conductances known to be compensated in small circuits (12). The infinitesimal five-mechanism curvature of §1.4 and this finite per-channel elasticity are distinct measures (an infinitesimal whole-class scaling versus a per-channel finite elasticity); the elasticity is the behaviourally meaningful one and is reported as primary. File `baai_perchannel_hessian.json`.

4.3 Full-connectome manifold probe (S5.3)

Over the full 3076 connection weights (1992 synaptic + 1084 gap-junction) we perturb every weight individually and assemble the Gauss–Newton curvature against 9600 motor-output observables, so the rank is not limited by the measurement. The effective dimension is 8 at 90% and 47 at 99%, participation ratio 2.55: 3068 of the 3076 directions are flat. Only 4 weights produce no measurable change, so the flat directions are degeneracy rather than unconnected parameters. Per-connection elasticities span seven orders of magnitude (first percentile 6.5×10^{-6} , ninety-ninth 59.7). Splitting by connection type, chemical synapses carry 87% of the curvature and gap junctions 13%, from 1992 and 1084 connections respectively, so per connection a chemical synapse contributes about five times more. The degeneracy of this model therefore lies principally in the connectome, not only in the single-cell biophysics. The measurement is not a property of the published parameter point: displacing the entire weight vector to four new operating points ($\theta_i e^{\mathcal{U}(-\ln f, \ln f)}$ per connection, two points each at $f=1.25$ and $f=1.5$) and repeating the probe over an identical 300-connection subsample gives effective dimension 8–12 (90%) / 28–30 (99%) at every point, with the spectrum spanning 14 orders of magnitude throughout, against 9/29 at the published point on the same subsample. The stiff subspaces moreover coincide across points: the mean pairwise principal cosine is 0.89 (range 0.78–1.00), where randomly oriented subspaces of the same dimension would give 0.145 ± 0.012 . The low dimension, and the identity of the constrained directions, are properties of the region of parameter space the model occupies. Files `E42_baaiworm_connectome.json`, `E58_baai_multipoint.json`.

4.4 Behaviour-based prediction of the internal neural state (S5.4)

Rolling out BAAIWorm (NEURON (13)) with behaviour-recovered parameters and reading the internal membrane voltage of all 136 neurons, the generated neural dynamics are low-dimensional (PCA participation dimension ≈ 4.7 –4.8) with mean absolute pairwise correlation 0.37–0.39, the same regime as the real whole-brain calcium recordings of Kato et al. (14) (mean absolute correlation 0.21). Behaviour-only parameter generation thus reproduces the gross statistics of the unobserved neural state, though the recovered and default arms are similar and the comparison is qualitative (membrane voltage versus calcium fluorescence; behaving versus immobilised). Files `_b2_stats_*.json`.

4.5 Mutant mechanism localisation (S5.5)

Knocking each named mechanism to $0.4\times$ and asking whether the resulting behavioural-shift signature identifies the perturbed mechanism: the sloppy calcium channels are behaviourally near-silent (Fig. 1b) (signature norm EGL-19 0.07, UNC-2 0.12, SLO-1/SLO-2 0.09–0.16) and so cannot be localised from behaviour, while the detectable mechanisms (synaptic, passive, motor) are mutually confusable (cosine > 0.9 with several others). Behaviour alone does not uniquely localise the perturbed mechanism, which is the mechanism-level expression of the behavioural-equivalence manifold. File `baai_b3_localise.json`.

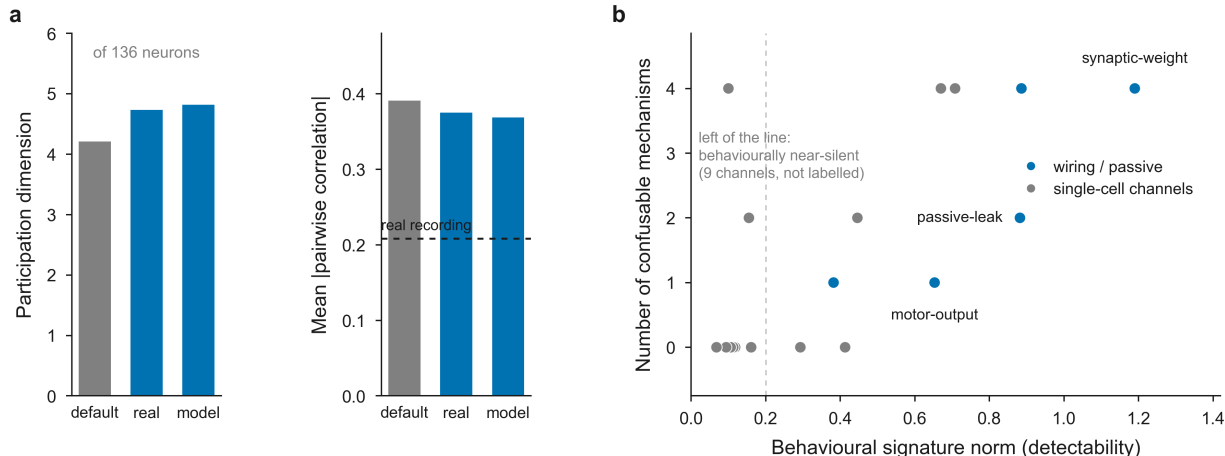


Figure 1: Internal neural state under behaviour-recovered parameters, and the behavioural detectability of single-mechanism knockdowns. **a** Mean absolute pairwise correlation of the internal membrane-voltage dynamics of all 136 BAAIWorm neurons, under default parameters and under parameters recovered from behaviour alone against a real (“real”) or model-generated (“model”) target. Dashed line: the same quantity computed from the real whole-brain calcium recordings of Kato et al. (14). The generated dynamics are low-dimensional (PCA participation dimension ≈ 4.7 – 4.8) with mean absolute correlation 0.37 – 0.39 , against 0.21 in the recordings. **b** Behavioural detectability of an in-silico knockdown of each named mechanism to $0.4\times$, plotted against the number of other mechanisms whose behavioural signature it is confusable with. Blue: wiring and passive mechanisms; grey: single-cell channels; only the three mechanisms named in the text are labelled. Dashed line: the detectability threshold below which a knockdown is behaviourally near-silent. The fast calcium and calcium-activated potassium channels fall below that threshold (signature norm 0.07 – 0.16) and so cannot be localised from behaviour at all, while the mechanisms that are detectable are mutually confusable (cosine > 0.9 with several others); behaviour alone therefore does not uniquely localise a perturbed mechanism. Comparison in **a** is qualitative: membrane voltage in a behaving model against calcium fluorescence in an immobilised animal.

4.6 External mutant-phenotype check (S5.6, null)

Comparing real gene-knockout behavioural-effect magnitude (Yemini OWMD, 43 strains) against the computed stiff/sloppy class and against gene conservation: the stiff-versus-sloppy contrast is not significant (Mann–Whitney $p=0.70$) and the effect–conservation correlation is weak (Spearman $\rho=0.33$, $p=0.42$, $n=8$ overlapping genes). This per-gene comparison probes the wrong granularity: the stiff/sloppy split is a property of coupled parameter *combinations* rather than single channels (8–10) (stiff/sloppy eigenvector participation $3.42/4.68$ of 7; §6.7), and homeostatic compensation lets individual conductances vary severalfold while the functional combination is preserved (12), so a single-gene knockout mixes stiff and sloppy components and need not rank with per-channel stiffness. We therefore read the stiff/sloppy geometry from the behavioural probes rather than from per-gene phenotype magnitudes. File `E3_mutant_phenotype.json`.

4.7 A fifth simulator: *Drosophila* connectome model (S5.7)

To extend the stiff/sloppy decomposition to a connectome-constrained *neural* model of the fly (the flybody/FlyGym musculoskeletal simulators have no conductance-level parameters), we add the

connectome-constrained optic-lobe network of Lappalainen et al. (15), built on the adult-*Drosophila* connectome (16, 17) (`flyvis`; 45,669 neurons, 734 free parameters in three named groups: cell-type biases, cell-type time constants, synapse-type strengths). On a synthetic drifting-grating stimulus, the finite-difference Hessian over the 65 per-cell-type biases read through 64 output channels has effective dimension 43 (90%) / 58 (99%) of 65 (participation ratio 36.3). That figure is a property of the assay, and this model is where the paper demonstrates it directly: reducing the recorded output channels from 64 to 32, 16, 8 and 6 moves the effective dimension to 24, 13, 7 and 5, close to one constrained dimension per channel, so the bias-only number tracks how much was recorded rather than the circuit. Measured across all three of the model’s free parameter groups, where sampled parameters (330) outnumber neither the observables nor the readout budget per parameter, the effective dimension is 14 (90%) / 32 (99%), the figure comparable with the other systems and the one the main text reports. The identity of the stiff axes does not depend on the assay: the stiffest cell types are the elementary motion detectors T4a–d and their medulla inputs (Tm4, Tm16), the sloppiest the photoreceptors (R5, R6) and lamina wide-field cells (Lawf1/2)—the motion-computing pathway is behaviourally load-bearing, the input gain is not. (We compare simulators by effective dimension rather than by a raw perturbation-sensitivity ratio: across the four motor models the latter ranges over three orders of magnitude with the observable and parameterisation.) Files `flyvis_b1.json`, `flyvis_manifold.json`.

4.8 Cross-system stiff/sloppy convergence (S5.8)

Applying the same Hessian-eigenspectrum probe to the canonical Prinz–Marder stomatogastric conductance population (a non-worm, non-fly crustacean circuit; §2.1) gives effective dimension 2.13/31 with stiff = leak and inhibitory-synapse conductances and sloppy = fast voltage-gated Na and delayed-rectifier K, the same biophysical split seen in the worm models, where the fast voltage-gated calcium channels (EGL-19, UNC-2) are sloppy (§5.2; Fig. 2). Across four real models and three phyla, behaviour/output is set by network wiring and slow/passive properties and left free along the fast voltage-gated inward currents. File `STG_b1analog.json`.

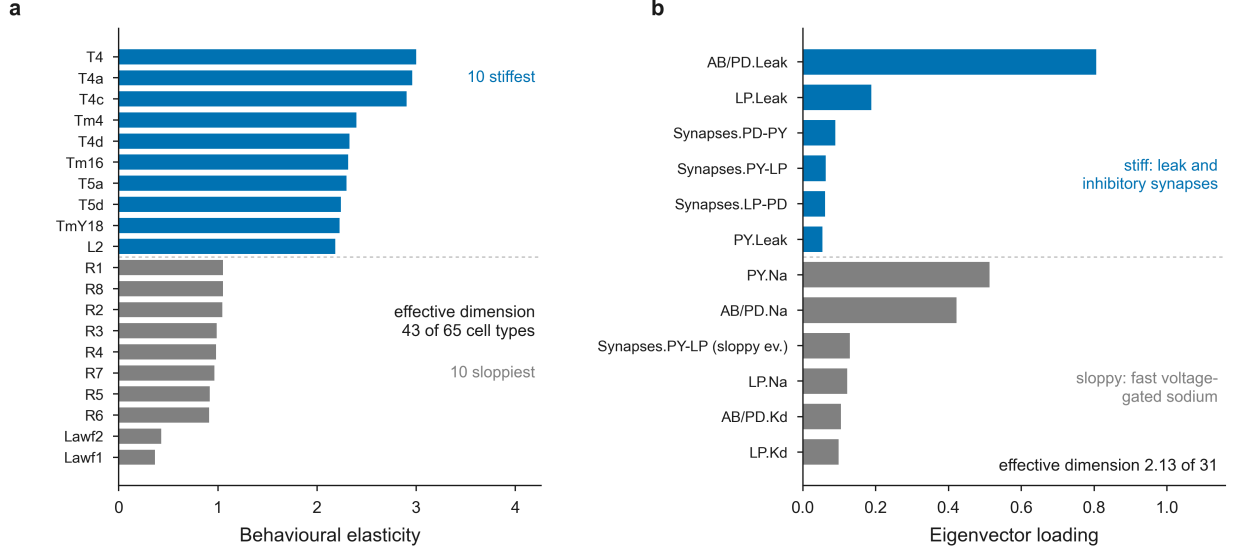


Figure 2: **The fly visual model measured by parameter group, and the crustacean pyloric population.** **a** Behavioural elasticity of the flyvis optic-lobe model by cell type, the ten stiffest (blue) and ten sloppiest (grey) of its 65 named types. The motion-detecting T4/T5 pathway and its medulla inputs are stiff; the photoreceptors and the lamina wide-field cells are sloppy. Which parameter group is probed matters as much as how many observables are recorded: the per-cell-type biases shown here give 43 of 65, the membrane time constants 5 of 65, the synaptic strengths 14 of 200, and the three groups measured jointly 14 of 330 sampled parameters, which is the figure comparable with the other systems here. **b** Eigenvector loadings of the Prinz–Marder pyloric conductance population under the same estimator, with the leak and inhibitory-synapse conductances stiff and the fast sodium conductances sloppy. This model is also the best conditioned in the study, with the response proportional to the perturbation across two decades of step size (ratio 0.498 for the biases). Curvature is the Gauss–Newton form $H = J^\top W J$.

5 The identifiable subspace: identity, tiling, and cross-species universality

5.1 Identification of the manifold with the eigenworm postural space (S6.1)

We reduce real *C. elegans* skeletons (OpenWorm Movement Database, N2; 8,432 frames) and model body postures to tangent-angle eigenmodes by PCA. Real posture has effective dimension 3.79 (49-point skeletons) / 4.14 (17-point), with four eigenworms capturing 0.96 / 0.94 of postural variance, reproducing the classical four-eigenworm decomposition (18). Two independently built worm models produce behaviour that lives in this same real eigenworm space: the modWorm posture subspace aligns with the real eigenworm basis at mean principal-angle $\cos 0.773$, and BAAIWorm at $\cos 0.68$ – 0.69 (67–72% of variance). Within each model it is the stiff mechanisms (synaptic, gap-junction, synaptic-rise/fall time) that move the eigenworm trajectory and the sloppy ones that do not (stiff/sloppy eigenworm-displacement ratio 1.17). The low-dimensional behavioural-equivalence manifold is therefore the neural control of the worm’s ~ 4 -dimensional posture. These alignments are significant against a null of random four-dimensional subspaces: drawing random $k=4$ subspaces in a d -dimensional posture ambient and computing the same mean principal-angle cosine, the modWorm value 0.773 has $p=0.0014$ and BAAIWorm 0.683 has $p=0.043$ even in a de-

liberately conservative $d=10$ ambient, falling to $p<10^{-4}$ at $d=20$ and $d=48$ (the full tangent-angle space); the random-subspace mean is 0.55/0.38/0.24 at $d=10/20/48$ (20,000 draws each). Files `EW_eigenworm.json`, `EW_baai_local.json`, `NCS_p1p5.json`.

5.2 Independence of the low dimension from observable resolution (S6.2)

Re-measuring the real modWorm behavioural Hessian with 7,200–14,400 posture observables (full body-segment angle and curvature time series, rather than six summary kinematics) leaves the effective dimension at 2 (90%) / 3 (99%) of 7, with spectrum $[1, 0.16, 0.013, 0.002, 0, \dots]$. The low dimensionality is a property of the behaviour, not of how coarsely it is sampled. File `RICH_observable.json`.

5.3 Complementary tiling: distinct behaviours constrain distinct axes (S6.3)

In BAAIWorm we contrast two behaviours—food-directed chemotaxis (food close; base net displacement 0.061) and plain locomotion in a weak gradient (food far; 0.231)—and perturb either the chemosensory parameters or the motor parameters. The chemosensory parameters are stiff during chemotaxis (behavioural sensitivity 0.872) but sloppy during locomotion (0.122): a $7.15\times$ context shift. The motor parameters show the opposite contrast (0.109 chemotaxis vs 0.357 locomotion, ratio 0.30). Each behaviour makes stiff the axis its function needs, and the two behaviours make *different* axes stiff, so the identifiable subspace is the union over behaviours, not the few directions of any one. File `BAAI_chemo.json`.

5.4 Union growth and saturation within one model (S6.4)

On the real modWorm model, the union effective dimension over a behaviour set grows with the number of behaviours and then saturates: across twelve behaviours the union reaches 4 (99%) / 2 (90%) of 7, leaving a residual 2–3 directions that no behaviour constrains. Four genuinely distinct locomotor presets (gentle/harsh $\times$ anterior/posterior touch) each constrain 2–4 of 7 mechanism directions, their leading stiff subspaces only partially overlapping (mean pairwise cos 0.54, range 0.35–0.77), and their union is 5 (99%) / 2 (90%). Quantifying the complementarity directly: the four behaviour-specific stiff directions span an effective 2.98 of 4 dimensions (a single shared direction would give 1; the participation dimension of their singular values), and each behaviour contributes a mean 0.535 component orthogonal to the span of the other three (per-behaviour 0.56/0.65/0.46/0.47). To establish that this growth is not the trivial consequence of non-identical subspaces, we compare the observed geometry against two explicit nulls, with 20,000 draws each. Under a *redundant* null, in which every behaviour constrains one shared direction, the span would be 1, the novel fraction 0 and the mean pairwise cosine 1. Under a *random-orientation* null, with directions drawn uniformly on the sphere in the same ambient space, the span is 3.50 (95% interval 3.00–3.88), the novel fraction 0.735 and the mean pairwise cosine 0.312. The observed values (span 2.978, novel fraction 0.535, cosine 0.525) lie between the two and are significantly more aligned than random ($p=0.020$ for the span, $p=0.082$ for the novel fraction, $p=0.016$ for the cosine). The behaviours therefore share substantial structure while each still contributes a distinct component. With only four behaviours, a bootstrap over behaviours is weak (resampling can draw duplicates and collapse the span), so we report the null comparison as the primary statistic and the bootstrap interval (span 1.45–2.98) only for completeness. File `null_model_tiling.json`. The residual core is genuine biological degeneracy, not an unused parameter (S6.7). Files `SAT_saturation.json`, `MB_behaviour_specificity.json`.

5.5 Cross-species: the same growth structure across three phyla (S6.5)

The same union construction applied to two further simulators shows the identifiable dimension scaling with behavioural richness in each. *Fly larva* (larvaworld; 10 behavioural-module parameters): four genuinely different sensory behaviours (free exploration, chemo-orbiting, anemotaxis, feeding) each have single-behaviour effective dimension 3–5 of 10 and engage different modules (olfactory, wind-sensory, feeding); their union is 8 (99%) / 4 (90%), roughly double the single-behaviour value. *Mammal* (Virtual Rodent (19, 20), `dm_control` (21), MuJoCo (22); 38 actuators): a six-behaviour repertoire drives a still-rising union curve ($3.67 \rightarrow 5.93 \rightarrow 7.67 \rightarrow 9.27 \rightarrow 10.5 \rightarrow 12.0$). The absolute counts are not comparable across species: the three models are parameterised differently and the rodent uses open-loop sinusoidal drives rather than trained natural behaviours, so we report the invariant structure (per-behaviour low-dimensional, complementary across behaviours, union set by repertoire), not a cross-species numerical law. Files `LARVA_manifold.json`, `RODENT_manifold.json`.

5.6 Concordance of two independently constructed worm models (S6.6)

Probed identically (per-mechanism Gauss–Newton elasticity, the same six observables, $\delta=0.25$), `modWorm` (coupled-ODE, 7 mechanisms) and `BAAIWorm` (NEURON, 20 named channels) agree at the class level: the network-wiring axes (synaptic, gap-junction) carry a stiff fraction of 0.833 (`modWorm`) / 0.868 (`BAAIWorm`), the single-cell biophysical axes only 0.083 / 0.158. “Wiring stiff, single-cell sloppy” is thus a property of the worm, not of one implementation. (The fine ordering *within* the stiff classes differs between the two models (`modWorm` ranks synaptic highest, `BAAIWorm` the passive leak), so the rank concordance over $n=3$ shared classes is not used as evidence; only the class-level split is.) File `G2_crosssim_align.json`.

5.7 Sloppy directions as coupled parameter combinations (S6.7)

On the real `modWorm` model the stiff and sloppy Hessian eigenvectors are both distributed parameter combinations: stiff-eigenvector participation 3.42, sloppy 4.68 of 7. Moving a fixed step along the sloppy eigenvector changes behaviour less than moving along the stiff eigenvector or along a single parameter (stiff/sloppy behavioural-change ratio 0.74; single-parameter/sloppy 0.97 at step 0.6), confirming the sloppy direction is a behaviour-preserving coupled combination rather than a parameter with no effect. The residual degenerate core is therefore a true biological neutral space. File `CP_coupling.json`.

5.8 Robustness of the effective-dimension measurement (S6.8)

Three controls establish that the low effective dimension is neither a single-point nor a method artefact. (i) *Multiple manifold points*: at five independent points on the real `modWorm` manifold the per-mechanism Hessian effective dimension is $[4, 3, 1, 3, 4]$ of 7: low everywhere, not at one optimum. A denser probe at 24 random points drawn in the same prior region ($\sigma=0.15$) gives 90% effective dimension 1–3 and 99% dimension 2–5 at every point, so the low dimension is a property of the region rather than of the published parameter set; the mechanism ordering correlates across points at Spearman 0.63 ± 0.19 and the identity of the single stiffest mechanism varies (shared by 38% of point pairs), so the leading direction rotates within a stiff subspace that itself stays low-dimensional. File `E7_E11_modworm.json`. (ii) *Ruler calibration*: on constructed spectra of known rank the estimator is exact (flat- k spectra recover $k=2, 5, 10$; uniform- d recover $d=8, 20$; an end-to-end sloppy geometric spectrum recovers 2). (iii) *Statistics of probe agreement*: the Hessian-stiffness versus manifold-drift

ranking agree at Spearman $\rho=0.964$, exact permutation $p=0.0028$ (5040 permutations), bootstrap 95% CI [0.565, 1.00]. Files `mw_D1R6.json`, `D2_ruler_control.json`, `D3_modworm_stats.json`.

5.9 Robustness of the low dimension to the spectral-mass threshold (S6.9)

The effective dimension is the number of eigenvalues reaching a chosen fraction of the spectral mass; the main text reports the 90% and 99% values. Recomputing across thresholds 80/90/95/99%, and alongside the threshold-free participation ratio ($PR = (\sum \lambda)^2 / \sum \lambda^2$), the low-dimensional conclusion is insensitive to the choice (Table 2): every system stays low-dimensional at all thresholds, and the participation ratio—which uses no threshold—is ≤ 3.4 throughout. The rodent union is the highest, exactly as the behavioural-repertoire law predicts (a richer repertoire constrains more directions), not because of a threshold artefact.

system	dim	effective dimension at threshold				PR
		80%	90%	95%	99%	
modWorm full Hessian	30	1	1	1	1	1.00
modWorm 7-mechanism	7	2	3	3	4	2.23
modWorm rich-observable	7	1	2	2	3	1.35
BAAIWorm per-channel	20	1	2	2	3	1.34
larva union (4 behaviours)	10	3	4	5	8	2.07
rodent union (6 behaviours)	38	4	6	7	≥ 10	3.35

Table 2: **Effective dimension across spectral-mass thresholds and by the threshold-free participation ratio.** Effective dimension at a threshold q is the smallest number of eigenvalues whose cumulative magnitude reaches a fraction q of the total spectral mass; the participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ uses no threshold at all. Every system remains low-dimensional at all four thresholds and by the threshold-free measure, so the conclusion does not depend on where the cut is placed; the rodent union is the largest, as the behavioural-repertoire result predicts. The rodent row uses the stored top-ten union spectrum, so its 99% entry is a lower bound; the full-spectrum union is 12 (§6.5). Curvature is the positive semi-definite Gauss–Newton form in every row.

6 Reproducibility

All external datasets are public: Prinz–Marder STG model population (7); Allen Cell Types; WormBase ParaSite / Ensembl Metazoa; OpenWorm Movement Database. All simulators are the published releases cited in the main text, run at the settings given above. Each result section names its saved result file; every reported quantity and figure is regenerated from those files by the accompanying reproduction code.

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7 Ground-truth verification at the cell scale

7.1 Panel, protocol and estimator (S7.1)

Eleven published cell models—five neuronal (Hodgkin–Huxley (23), Hodgkin–Huxley with an A-type current (24), Wang–Buzsáki (25), Morris–Lecar (26), FitzHugh–Nagumo (27)) and six intracellular oscillators (Kholodenko MAPK (28), repressilator (29), Goodwin (30), Selkov (31), Brusselator (32), Goldbeter mitotic (33))—are implemented directly from their published equations and verified against the published dynamical structure: the two models carrying the main claims agree with independent second transcriptions to within 0.1%, and each of the others reproduces a structural signature of its source (the Brusselator Hopf threshold at $B = 1 + A^2$, the Goodwin requirement of a Hill coefficient above eight, the one-third-period phase symmetry of the repressilator, the Sel’kov and FitzHugh–Nagumo oscillation windows, the Goldbeter period of 25 minutes, fast spiking at 60 Hz in the Wang–Buzsáki neuron). They are measured with the identical Gauss–Newton estimator used throughout, in log-parameter space, with the observable count pinned to a fixed multiple of the parameter count so no model is read through fewer observables than parameters. Recovery follows the organism-scale protocol: the target is the published parameter set and the search starts from a random point $\theta_i^{\text{init}} = \theta_i^* e^{\eta_i}$, $\eta_i \sim \mathcal{U}(-\ln f, \ln f)$, drawn independently of the truth, at $f = 1.5, 3$ (the organism-scale regime) and 10. The loss is built from phase-invariant features of the settled trace (amplitude quantiles, spectral magnitude, autocorrelation), computed at full resolution and only then reduced, which keeps observables in excess of parameters while removing the phase sensitivity that makes pointwise trajectory losses untraversable for oscillators: pointwise losses converged on 11% of the same starts at $f=3$ and on none at $f=10$, against 47% and 36% for the feature loss. The discarded transient fraction is chosen per model by stationarity of the settled trace (quarter-range agreement), which matters only for the A-current neuron, whose default window straddles a silent stretch. A fit counts as converged when the behavioural distance falls below 0.02; only converged fits contribute parameter errors. Files `pgob_cell_panel13_result.json`, `pgob_cell_panel13b_result.json`.

7.2 Both ends of the geometry, against truth (S7.2)

The 2–4-parameter neuronal models are recovered exactly: Hodgkin–Huxley converges on 72/72 random starts with median parameter error 6×10^{-5} , Morris–Lecar on 17/24 at 3×10^{-7} , the Wang–Buzsáki and A-current variants on 39/72 and 15/72 at $<3 \times 10^{-4}$ (their lower convergence rates reflect landscape difficulty, not degeneracy: every converged fit lands on the truth). The larger biochemical cascades are degenerate in the direct sense: MAPK matches the behaviour to 10^{-3} while its parameters end at median error 0.40, farther from the truth than the random start (0.31), with individual rate constants compensating each other by factors up to 8; Goodwin and the Goldbeter oscillator end at 0.10 and 0.17. Across the eleven models the converged parameter error rises with parameter count (Spearman $\rho=0.70$, $p=0.017$). The estimator itself is validated in both directions: it reports the canonical sloppy spectrum on the model family where sloppiness was described, and exact identifiability where recovery proves it (Fig. 4a,b).

7.3 Effective dimension is not recoverable dimension (S7.3)

On Hodgkin–Huxley, a grid of three parameter counts (3, 5, 8: conductances; + leak reversal and capacitance; + gating-rate scalings) $\times$ four observable sets (1 spike count, 7 F–I and spike-shape features, 200- and 1000-point voltage traces) gives a 90% effective dimension of 1 in all twelve cells, while the achieved recovery error spans 4% (three conductances against the F–I assay; g_{Na} , g_{K}

within 4–5%) to 38% (eight parameters against the same assay, where the leak and capacitance are absorbed by the gating-rate scalings) to 56% (Fig. 4c). The effective dimension measures how concentrated the curvature is, not how many parameters recovery can pin down; the trajectory-observable cells additionally fail for a different reason (the behaviour is never matched), which is a property of the loss landscape, not of identifiability. This grid also reconciles the exact single-neuron recoveries here with the conductance degeneracy reported for such models (7, 8): the two literatures occupy different corners of the same table, few parameters against a multi-condition assay versus many parameters against a single trace. Files `pgob_cell_ladder_result.json`, `pgob_cell_diag_result.json`.

7.4 Network size does not dilute the constrained fraction (S7.4)

Coupling N identical Hodgkin–Huxley cells through fixed sparse excitatory synapses, with three conductances free per cell and the observable budget pinned per parameter, the constrained fraction barely moves as N grows from 2 to 64 (Fig. 3) (6 to 192 parameters): 0.33 to 0.30 at a $10\times$ observable budget, with the effective dimension rising linearly at close to one constrained direction per cell throughout, while the eigenvalue spectrum widens from 2.1 to 4.3 decades. Size alone therefore deepens the degeneracy but does not dilute the constrained fraction, so the 0.26% constrained fraction of the connectome-scale model (§5.3) is not a generic consequence of its parameter count. File `pgob_cell_scale_result.json`.

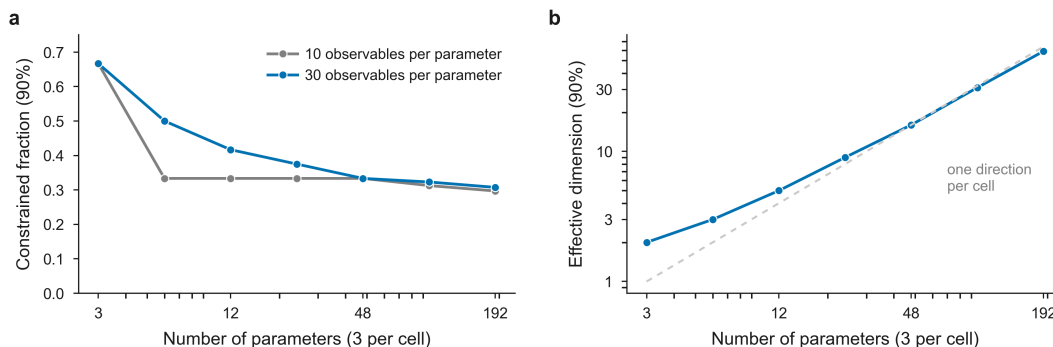
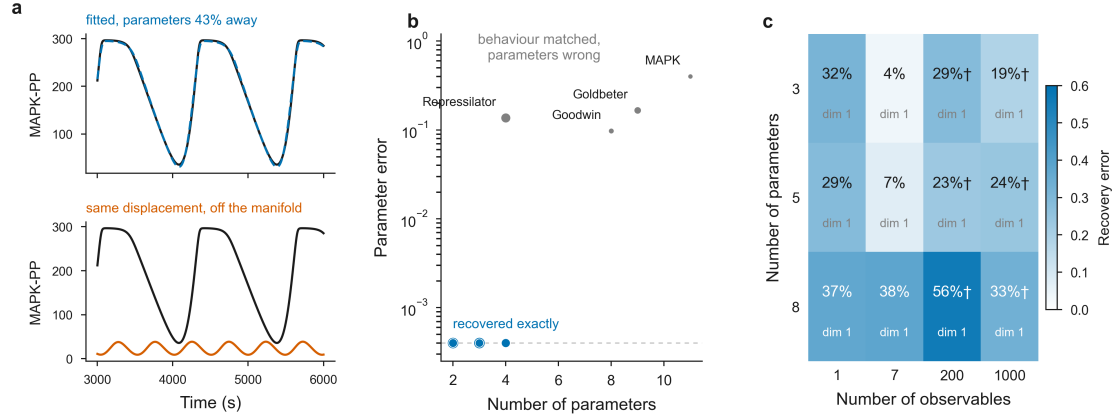


Figure 3: Network size deepens the degeneracy but does not dilute the constrained fraction. **a** Constrained fraction at 90% of the curvature as identical Hodgkin–Huxley cells are coupled into a fixed sparse network, three free conductances per cell, at two observable budgets. From 6 to 192 parameters the fraction moves from 0.33 to 0.30 at the $10\times$ budget and from 0.50 to 0.31 at the $30\times$ budget, two orders of magnitude away from the 0.26% of the connectome-scale model. **b** The effective dimension itself rises close to linearly, at about one constrained direction per cell (dashed reference), so the flat fraction is not a ceiling artefact; the widening eigenvalue spectrum (2.1 to 4.3 decades) deepens the degeneracy without diluting the fraction.



† behaviour never matched: the error reflects the optimiser, not identifiability

Figure 4: Ground-truth verification at the cell scale. **a** The degeneracy in the model’s own output. Top: the MAPK-PP oscillation at the true parameters (black) and at a fitted parameter set 43% away (blue, dashed), whose individual rate constants differ from the truth by factors of up to 8 yet whose behaviour is matched to 3%; the traces are indistinguishable. Bottom: a displacement of the same magnitude taken off the equivalence manifold (vermillion) changes the rhythm outright (period ratio 0.38). This fit sits just outside the 0.02 convergence gate and is shown for illustration; the quantitative claims rest on the gated fits of §7.2. **b** Median parameter error of behaviour-matched fits against parameter count, across eleven cell models recovered from random starts drawn independently of the truth (marker area $\propto$ convergence rate). Blue: models recovered exactly (error $< 10^{-3}$, on the dashed floor); grey: models whose behaviour is matched while the parameters remain wrong, labelled. **c** The same Hodgkin–Huxley model measured twelve ways: parameter count against observable count, cell colour and percentage giving the median recovery error, with the 90% effective dimension printed in each cell. The effective dimension is 1 everywhere while recovery spans 4–56%. Daggers mark cells in which the behaviour itself was never matched, so the error there reflects the optimiser rather than identifiability.